

Long Short -Term Memory-LSTM

ACUTE SHORT-TERM MEMORY LOSS



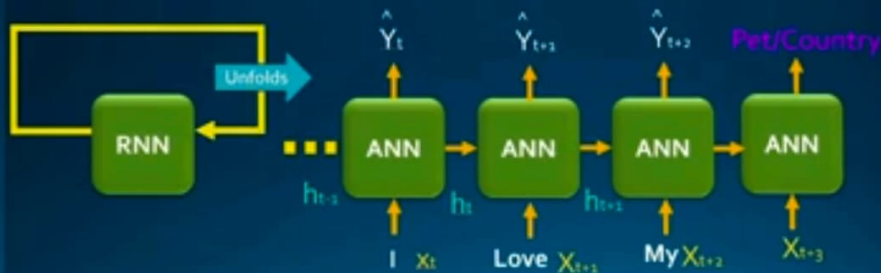
Ghajini (கஜினி)

$$\hat{y}_t = f(x_t, h_{t-1})$$

output input past memory

$$h_t = f_W(x_t, h_{t-1})$$

cell state function with weights W input old state



Ghajini Syndrome!!

$$1 \times 0.01 = 0.01$$

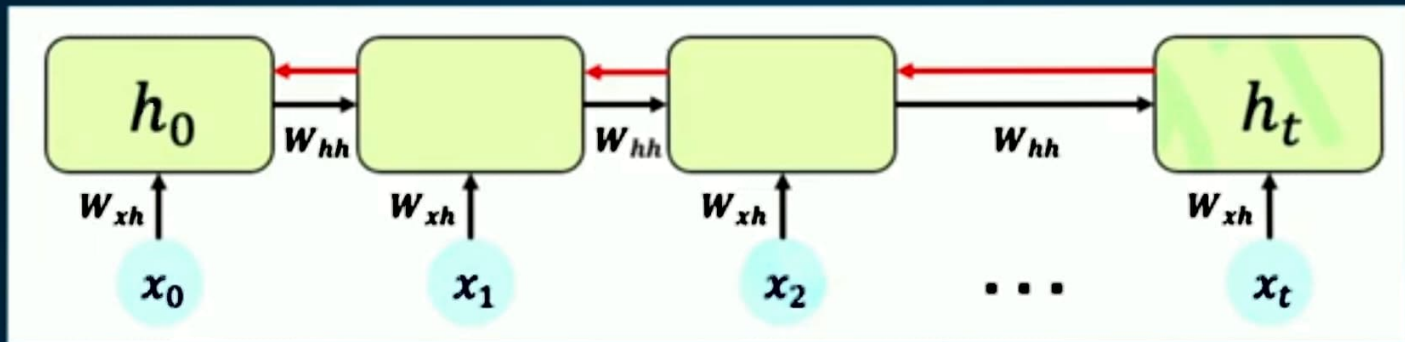
$$0.01 \times 0.01 = 0.0001$$

~ 0 (Vanishing)

$$1 \times 1000 = 1000$$

$$1000 \times 1000 = 1000000$$

$\sim \infty$ (Exploding)



Computing the gradient wrt h_0 involves many factors of W_{hh} + repeated gradient computation!

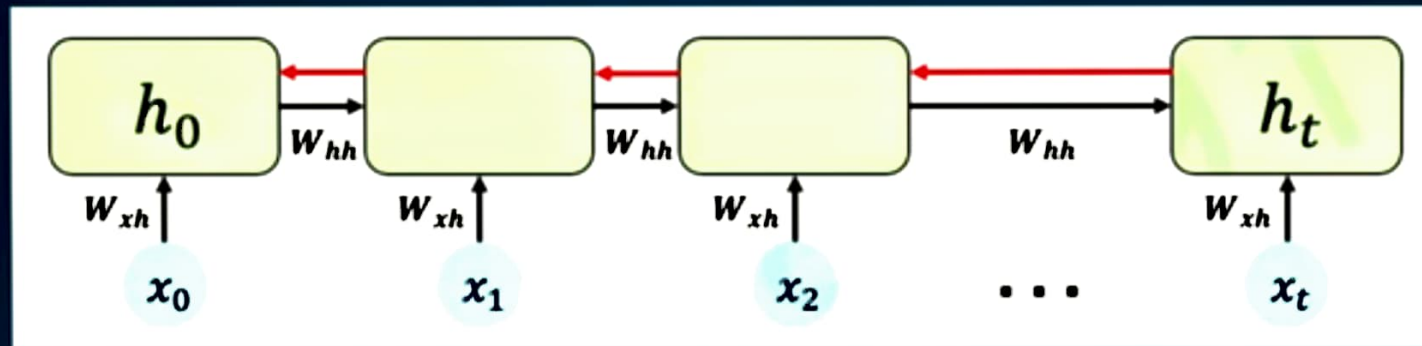
Many values > 1 :
exploding gradients

Gradient clipping to
scale big gradients

Many values < 1 :
vanishing gradients

1. Activation function
2. Weight initialization
3. Network architecture

Idea: use **gates** to selectively **add** or **remove** information within **each recurrent unit** with



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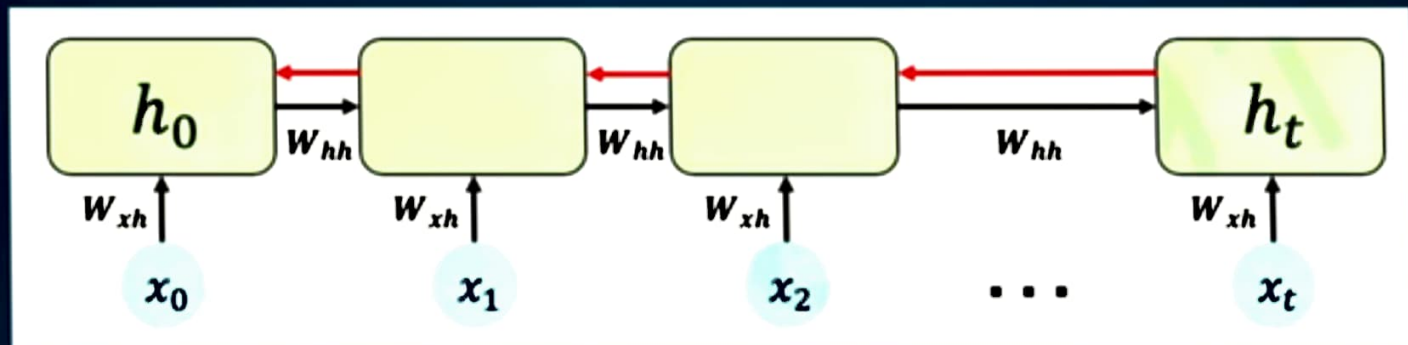
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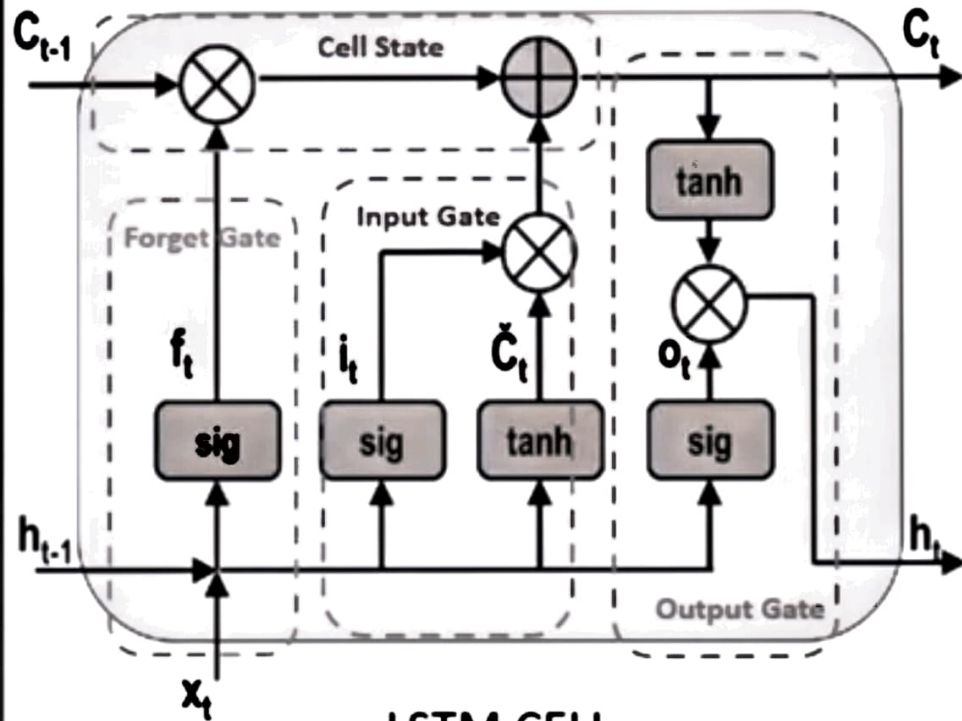
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Many values > 1

Many values < 1

Long Short Term Memory (LSTMs) networks rely on a gated cell to track information throughout many time steps.

3. Network architecture



LSTM CELL

sig = Sigmoid function

tanh = tanh function

$\otimes$ = point-by-point multiplication

$\oplus$ = point-by-point addition

$\rightarrow$ = vector connections

10AM

8PM

4PM

PAST (Memory)

FUTURE (Actions)

10AM

8PM

4PM

PAST (Memory)

FUTURE (Actions)

time

10AM

8PM

4PM

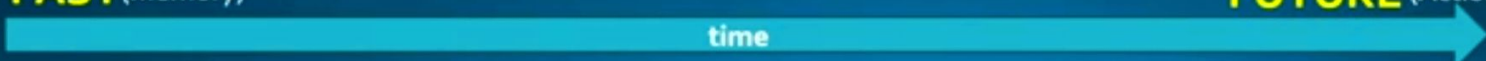
LONG TERM MEMORY

Short Term Memory

PAST (Memory)

FUTURE (Actions)

time





10AM

8PM

4PM

He had helped me

LONG TERM MEMORY

Short Term Memory

PAST (Memory)

FUTURE (Actions)

time



10AM

8PM

Memories



Forgot-keep updating

He had helped me

4PM

LONG TERM MEMORY

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LONG TERM MEMORY

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time



10AM



8PM

Memories



Forgot-keep updating

He had helped me

4PM

LONG TERM MEMORY

Arrange Money by Evening

Short Term Memory

PAST (Memory)

FUTURE (Actions)

time



10AM



4PM



8PM

Memories



Forgot-keep updating

He had helped me

He has cheated Dora

LONG TERM MEMORY

Arrange Money by Evening

Is he Trustworthy?

Short Term Memory

PAST (Memory)

FUTURE (Actions)

time



10AM



8PM

Memories



Forgot-keep updating

He had helped me

4PM

He has cheated Dora

LONG TERM MEMORY

Update



Arrange Money by Evening



Update

Is he Trustworthy?

Short Term Memory

PAST (Memory)

FUTURE (Actions)

time



10AM



8PM

Memories



Forgot-keep updating

He had helped me

4PM

He has cheated Dora

LONG TERM MEMORY



Update

Update



Arrange Money by Evening

Is he Trustworthy?

Borrow?

Short Term Memory

PAST (Memory)

FUTURE (Actions)

time



10AM



4PM



8PM

Memories



Forgot-keep updating

He had helped me

He has cheated Dora

LONG TERM MEMORY



Update

Arrange Money by Evening

Update



Is he Trustworthy?

Short Term Memory

Mix of long & short
Memories

Borrow?

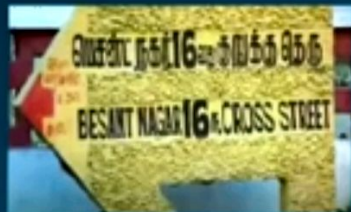
PAST (Memory)

FUTURE (Actions)

time



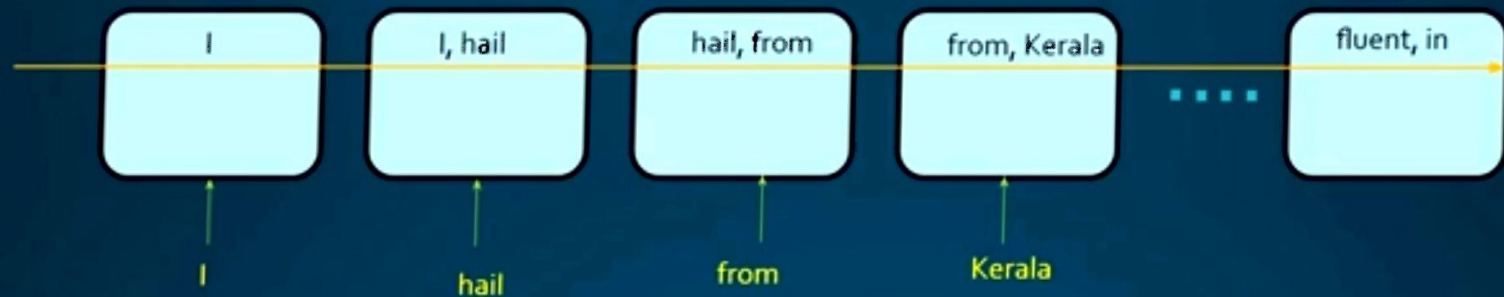
LONG TERM MEMORY



Short Term Memory

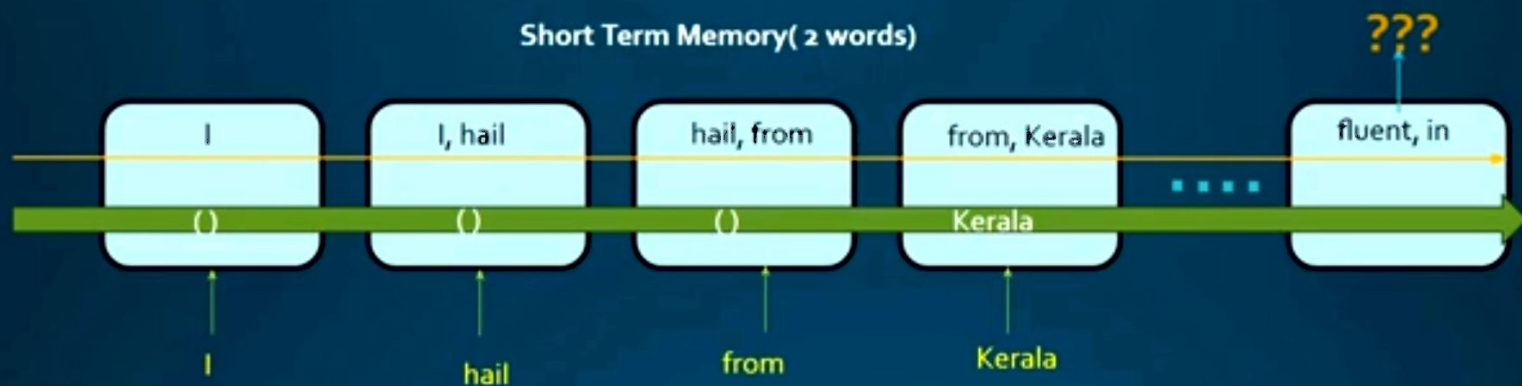
I hail from Kerala, often referred to as "God's Own Country." It's probably no surprise that I'm fluent in _____

Short Term Memory(2 words)



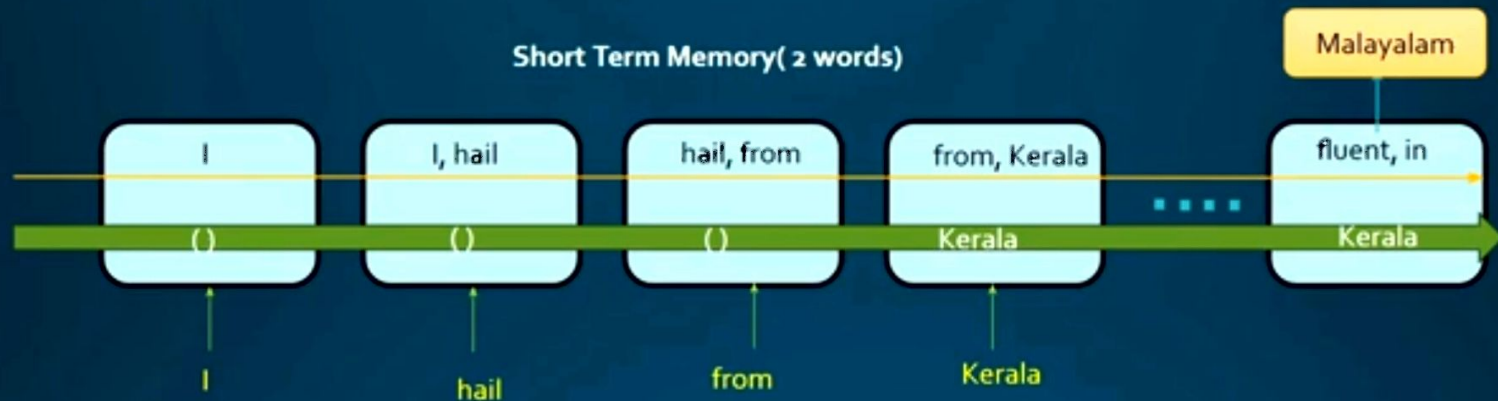
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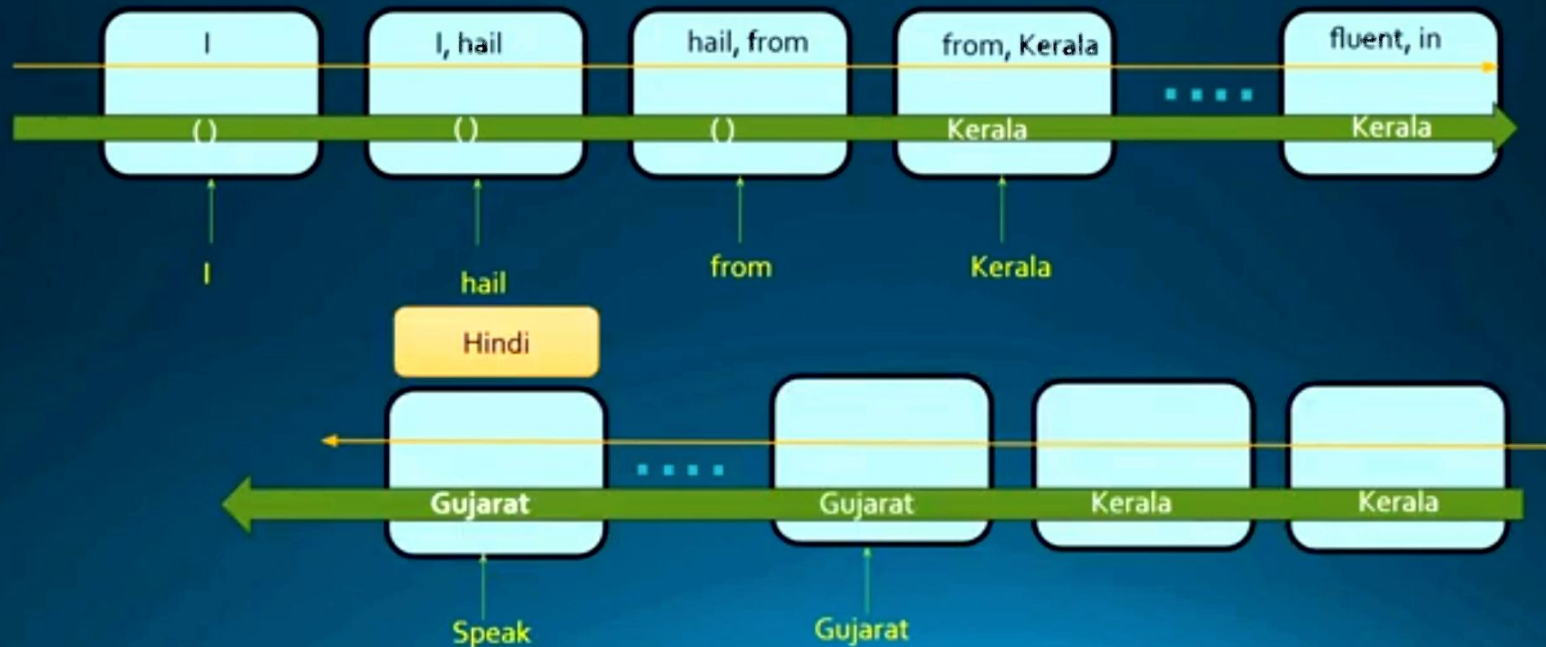
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Short Term Memory(2 words)



I hail from Kerala, often referred to as "God's Own Country." It's probably no surprise that I'm fluent in Malayalam. My brother has spent several years working in Gujarat, which is why he can speak _____ better than I can.

Short Term Memory(2 words)



c_{t-1} c_t f_t σ h_{t-1} h_t x_t

c_{t-1} c_t

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

 f_t σ h_{t-1} h_t x_t

c_{t-1}

Forgot about "Kerala"

 c_t

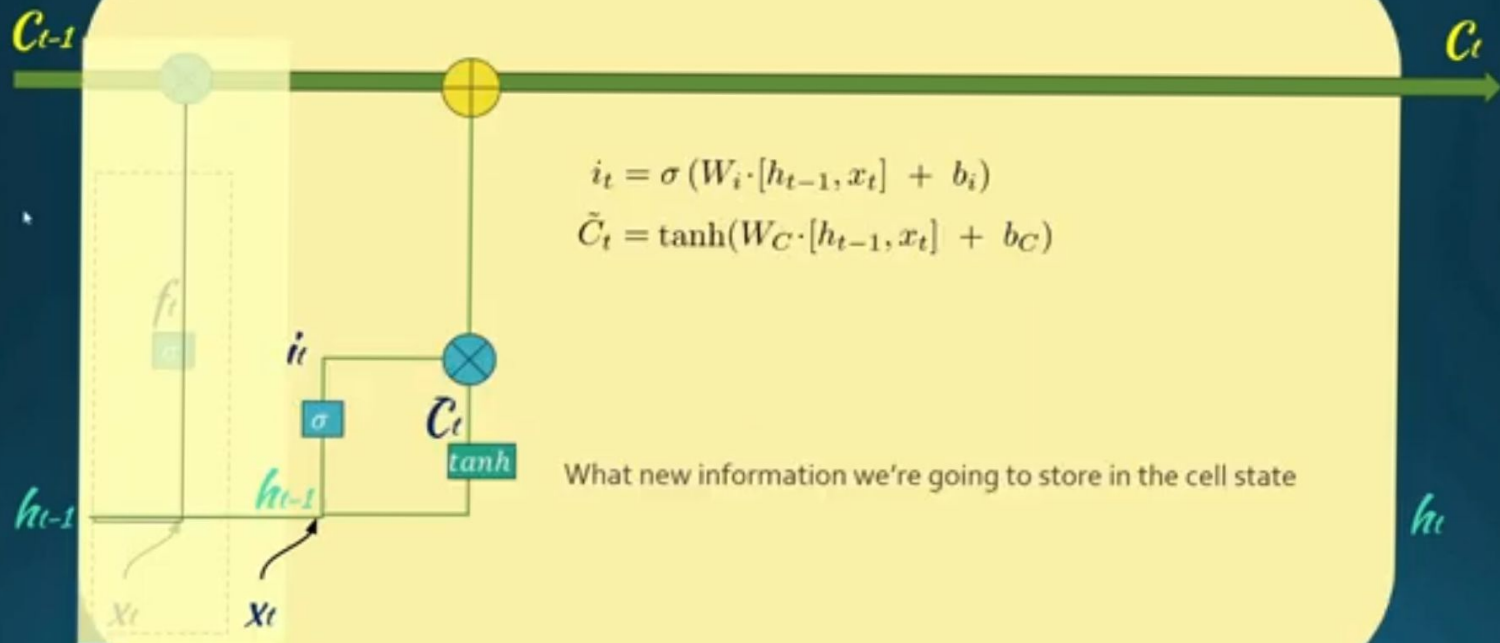
$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

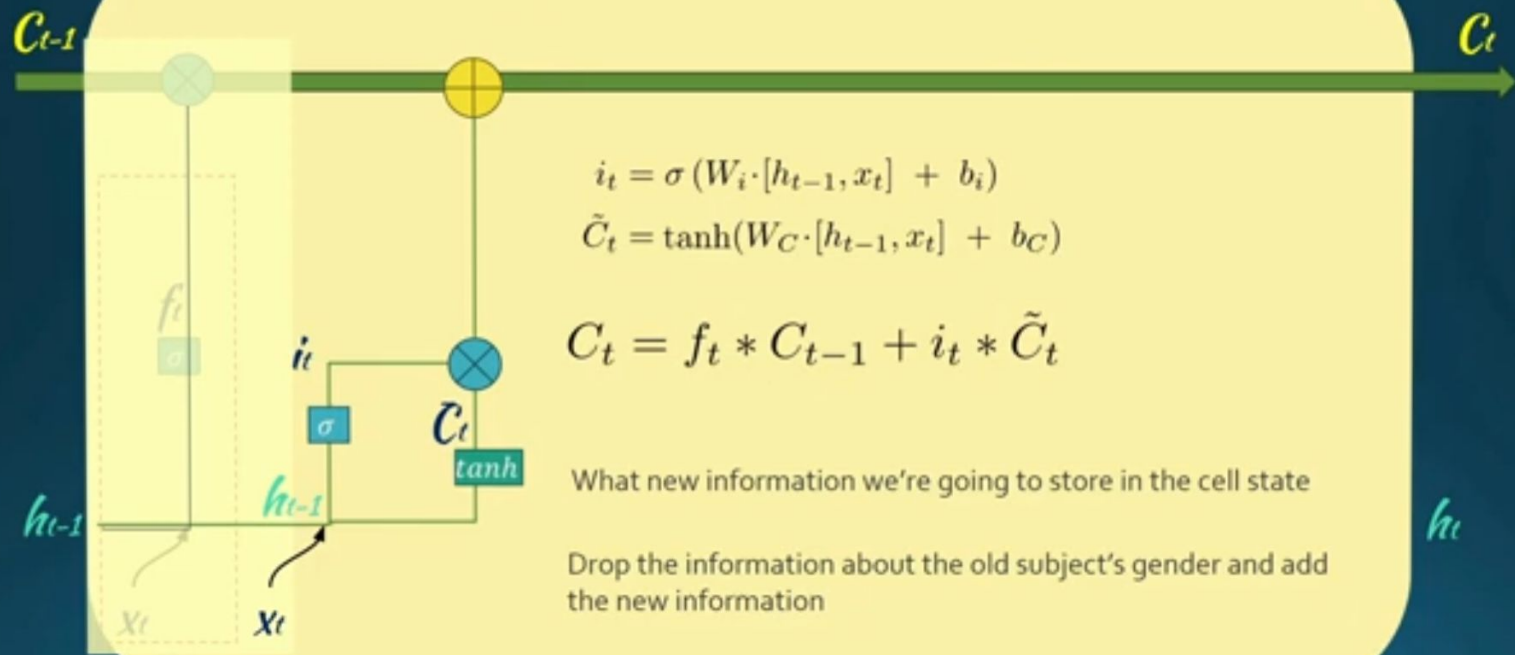
What information we're going to throw away from the cell state

 f_t σ h_{t-1} x_t

forget gate
layer

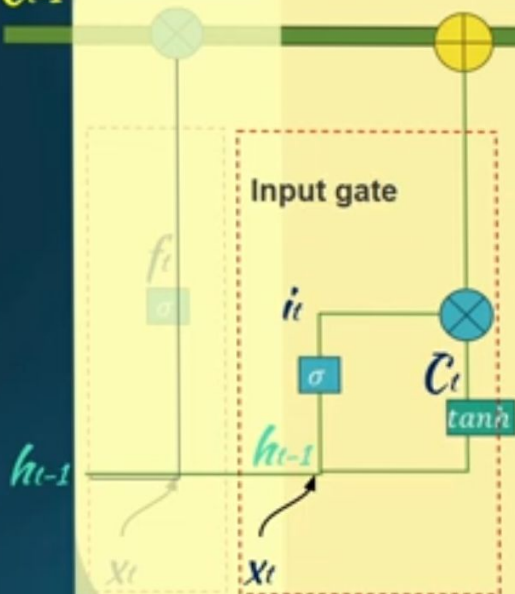
 h_t





C_{t-1}

Adding memory of "Gujarat"

 C_t 

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

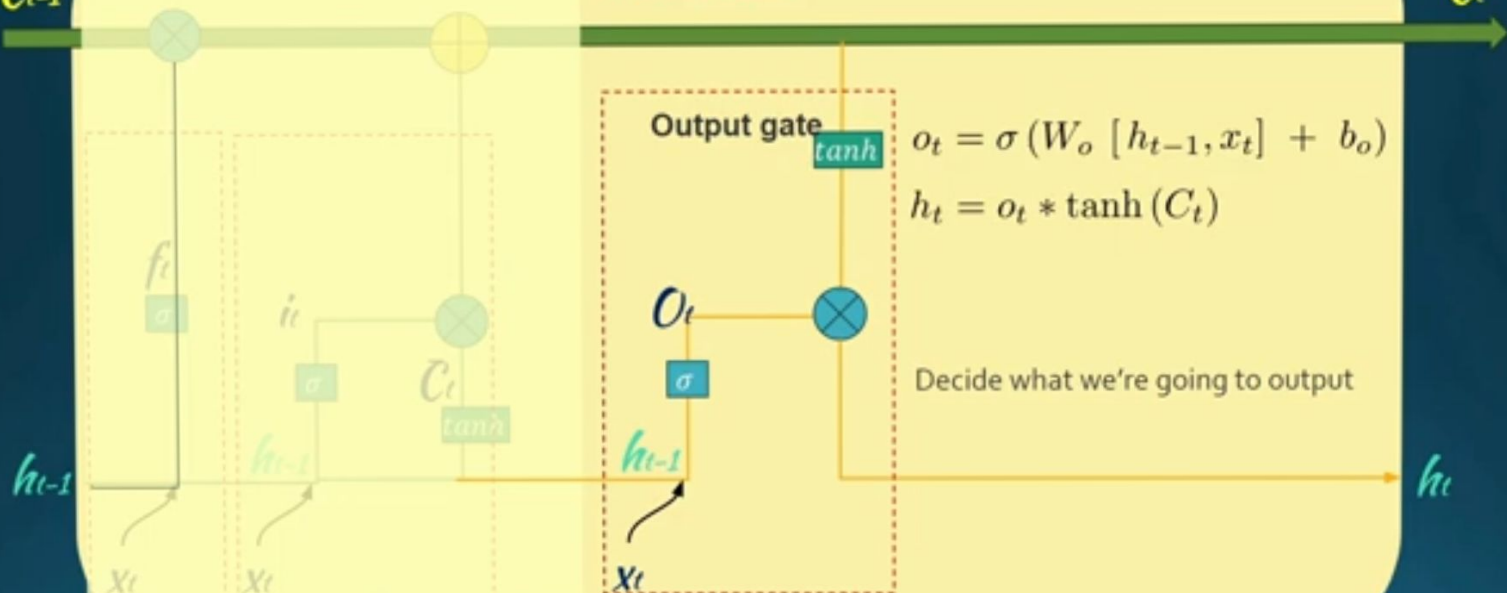
What new information we're going to store in the cell state

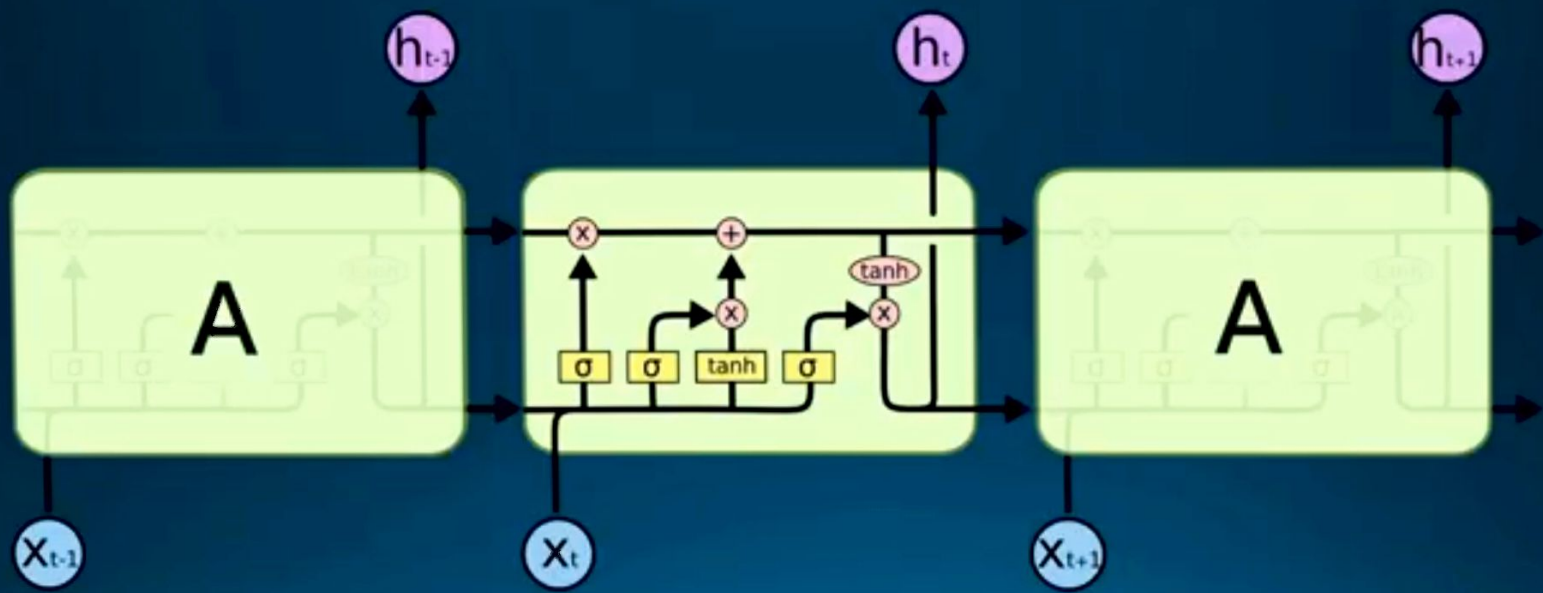
Drop the information about the old subject's gender and add the new information

 h_t

C_{t-1}

Predict "Hindi"

 C_t  h_t



The repeating module in an LSTM contains four interacting layers